

★ When 2 lines are Parallel and cut by a Transversal

1. Alternate Interior Angles Theorem

angles are =

$$m\angle 3 = m\angle 6$$

2. Alternate Exterior Angles Theorem

angles are =

$$m\angle 1 = m\angle 8$$

3. Corresponding Angles Theorem

angles are =

$$m\angle 1 = m\angle 5$$

4. Consecutive Interior Angles (same-side interior) Theorem

angles are supplementary

$$m\angle 3 + m\angle 5 = 180^\circ$$

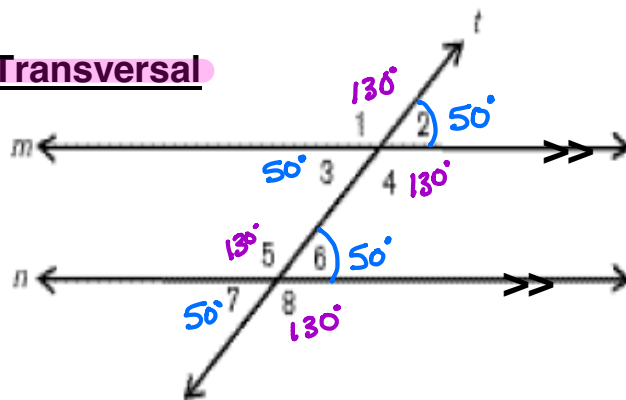
$$m\angle 4 + m\angle 6 = 180^\circ$$

5. Consecutive Exterior Angles (same-side exterior) Theorem

angles are supplementary

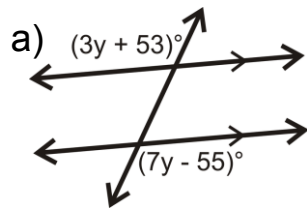
$$m\angle 1 + m\angle 7 = 180^\circ$$

$$m\angle 2 + m\angle 8 = 180^\circ$$



Example: For a and b, identify the angles. Find the value of the variables.

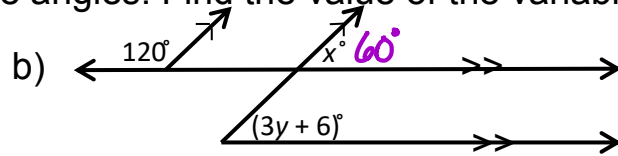
Alternate
Exterior
Angles



$$3y + 53 = 7y - 55$$

$$\boxed{27 = y}$$

x is
consecutive
exterior
with 120°



$$x + 120 = 180$$

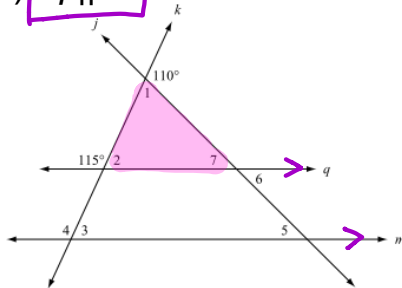
$$\boxed{x = 60}$$

Corresponding
Angles

$$60 = 3y + 6$$

$$\boxed{y = 18}$$

c) $\boxed{q \parallel m}$. Find the measure of the angles.



$$m\angle 1 = \underline{70^\circ}$$

$$m\angle 5 = \underline{45^\circ}$$

$$m\angle 2 = \underline{65^\circ}$$

$$m\angle 6 = \underline{45^\circ}$$

$$m\angle 3 = \underline{65^\circ}$$

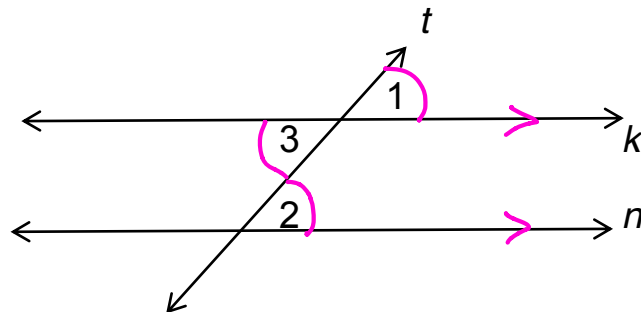
$$m\angle 7 = \underline{45^\circ}$$

$$m\angle 4 = \underline{115^\circ}$$

Proof

Given: $k \parallel n$

Prove: $\angle 1 \cong \angle 2$



Statements	Reasons
① $k \parallel n$	① given
② $m\angle 1 = m\angle 3$	② Vertical Angle Theorem
③ $m\angle 2 = m\angle 3$	③ Alternate Interior Angle Thm.
④ $m\angle 1 = m\angle 2$	④ Transitive Prop. of =
⑤ $\angle 1 \cong \angle 2$	⑤ Def. of $\cong$ Angles

